

Information Analysis of ENCA Summary Statistics

Mutual information between learned summaries, parameters and noise in a stochastic double-well model



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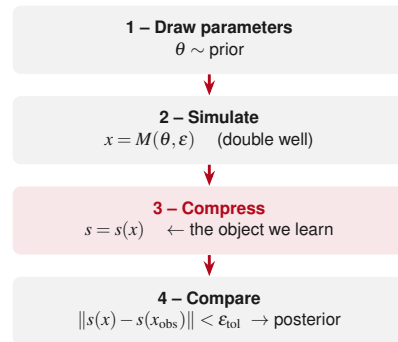
Why learn summary statistics?



The problem. Many stochastic models can be simulated but have no explicit likelihood. Inference is then simulation-based (ABC): simulate, compare with the data, accept or reject.

The comparison uses **summary statistics** $s(x)$: a few numbers per series that must keep the information about the parameters and ignore the randomness.

Hand-crafting them is hard. The ENCA (Albert et al. 2022) learns them with an autoencoder.



Objective: before the solar dynamo, check on a simple, fully controlled model that the compression works, using information theory.

1 Estimate $MI(X, \theta)$, $MI(X, \varepsilon)$, $MI(S, \theta)$, $MI(S, \varepsilon)$ with three estimators

2 Check $MI(S, \theta) \approx MI(X, \theta)$ and $MI(S, \varepsilon) \approx 0$

3 Change one network knob: does more information mean better inference?

4 Compare with a plain VAE (Kingma–Welling) of the same latent size

Plan: model › ENCA › results › MI estimators › the four MIs › filter sweep › VAE comparison › $q > p$

Sufficiency

The statistics keep (almost) all the information the series carries about the parameters:

$$MI(S, \theta) \approx MI(X, \theta)$$

S is computed from X , so it can never be larger: sufficiency means reaching the ceiling.

Concentration

Two runs with the same θ give similar statistics: the statistics ignore the noise.

$$MI(S, \varepsilon) \approx 0$$

In the paper: small $H(S|\theta)$.

How many? $q = p + 2$

One number per parameter is not enough: runs with the same θ still differ; for example, in which well is occupied.

We add two free dimensions.

Why it matters here. In a bistable model the same θ can produce different behaviours. The statistics must also record which behaviour occurred, and this cannot be stored in the parameter estimators. We check both criteria and the role of the free dimensions with mutual information.

$$x_{t+1} = x_t + \Delta t (ax_t - x_t^3) + \sigma \sqrt{\Delta t} \, \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

overdamped Langevin dynamics in the potential $V(x) = -ax^2/2 + x^4/4 \implies$ two wells at $\pm\sqrt{a}$

a

Well separation: *where* the trajectory sits

σ

Noise amplitude: *how much* it jitters

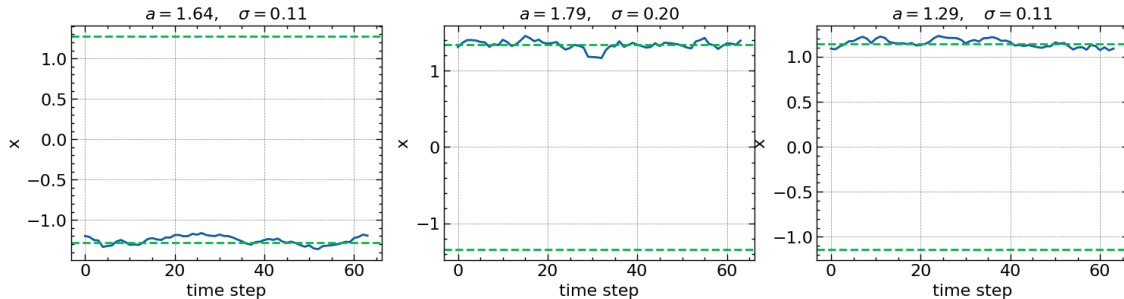
ε

Bare noise, drawn outside the simulator

Why this model. Simple and fully controlled and bistable: which well is reached is decided by the noise during a discarded warm-up, not by θ . The same $q > p$ behaviour as the solar dynamo, in miniature.

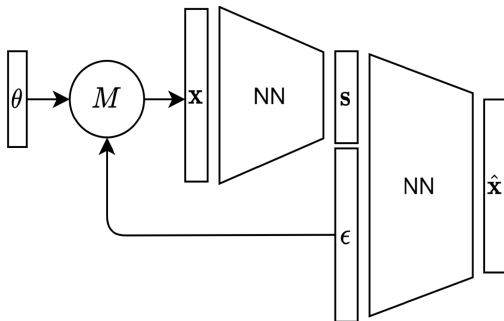
Priors $a \sim U(0.5, 2)$, $\sigma \sim U(0.05, 0.3)$ · 64 observed steps after 64 warm-up steps · 3000 training / 800 test series

What the data look like



three training trajectories: wells at $\pm\sqrt{a}$ (dashed), jitter set by σ

Each series stays in one well for the whole observed window. **The level tells us a , the jitter tells us σ , and which well it is was decided by the warm-up noise.**



ENCA architecture [Albert et al. 2022]

Key idea

The decoder receives the **exact noise** ϵ used by the simulator, so encoding the noise is useless.

The encoder can only improve the reconstruction by passing **parameter information**: s becomes a minimal set of summary statistics.

$s \in \mathbb{R}^4 = 2$ parameter estimators + 2 free dimensions

$$L = \text{MSE}(\hat{x}, x) + \lambda \text{MSE}(s_{1..p}, \theta), \quad \lambda = 2$$

The first $p = 2$ summaries become parameter estimators; the other two are free.

Encoder

1D CNN on two channels (the series x and its increments Δx) with **mean + std pooling** over time.

Decoder

Two bidirectional LSTM layers, fed the tiled summaries and the bare noise ε .

$\approx 12\text{k}$ parameters in total; trained in minutes on a laptop.

a is a level, σ is a spread. a sets *where* the series sits: a time average reads it immediately. σ sets *how much* it jitters, and an average destroys a spread.

Two design choices of our encoder:

1. a second input channel with the increments
 $\Delta x_t = x_t - x_{t-1}$, whose spread is essentially $\sigma\sqrt{\Delta t}$;
2. global pooling of the *standard deviation* over time, not only of the mean.

The same idea as the paper's estimator of σ for its bistable map, built from the step-to-step residuals $x_n - \alpha f(x_{n-1})$.

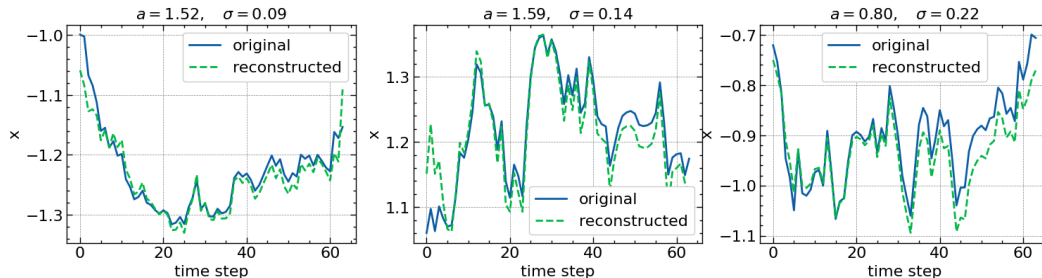
small σ : smooth



large σ : jittery



same mean (dashed), different spread:
the mean alone cannot tell them apart



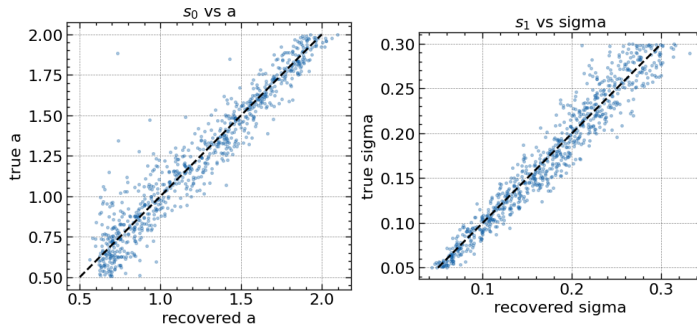
original vs reconstruction from (s, ϵ) ; three test series

Given the summaries and the bare noise, the decoder rebuilds the series: **test MSE** ≈ 0.003 on normalised data (0.3% of the variance). It tracks the occupied well, the level and the jitter.

64 $\rightarrow$ 4

each series compressed
into its summaries

Parameter recovery: it works



Both parameters sit on the $y = x$ line:

$$R^2(a) = 0.92$$

$$R^2(\sigma) = 0.94$$

The first two summaries are usable parameter estimators, σ included.

first two summaries, de-normalised, against the true parameters (800 test series)

PART II

Did the compression keep *all* and *only* the information?

There are **two main problems** when dealing with $MI(X, \theta)$, $MI(X, \varepsilon)$, $MI(S, \theta)$, $MI(S, \varepsilon)$:

1

$MI(X, \varepsilon) \rightarrow \infty$ because X is defined by a , σ and 64 values of $\varepsilon \implies$ **information given is huge**.

2

X and ε live in 64 dimensions. With 800 samples in 64-D, **no point has close neighbours**.

First, we **need MI estimators**, but we need densities. We never know the densities, only samples $\implies$ we need estimators that turn samples into a number.

1 • Binning

Discretise into bins and count

$$MI(X, Y) = H(X) + H(Y) - H(X, Y)$$

2 • Kernel density (KDE)

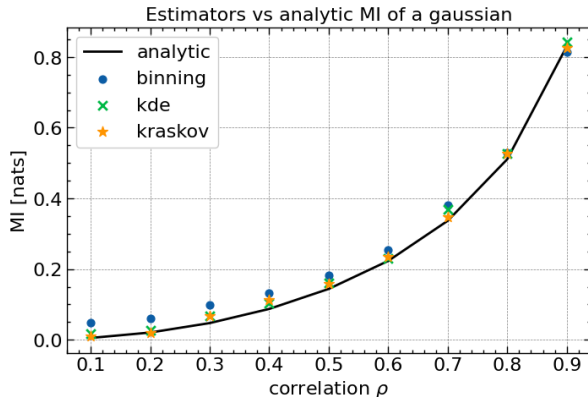
Rebuild the densities with Gaussian kernels

$$MI(X, Y) = \langle \log p(x, y) - \log p(x) - \log p(y) \rangle$$

3 • Kraskov k-NN

Count each sample's nearest neighbours in the joint and marginal spaces

$$MI(X, Y) = \psi(k) + \psi(N) - \langle \psi(n_x + 1) + \psi(n_y + 1) \rangle$$

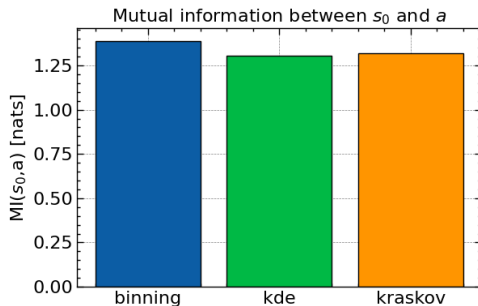


For two Gaussian variables with correlation ρ the mutual information is known exactly:

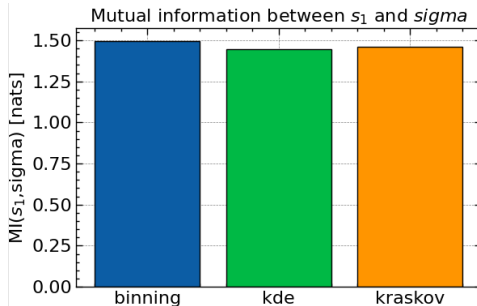
$$MI(X, Y) = -\frac{1}{2} \log(1 - \rho^2)$$

- ▶ All three estimators follow the analytic curve.
- ▶ Binning sits slightly above it at small ρ .

The three estimators on our data



One summary against one parameter (1-D vs 1-D):
the three estimators agree.



In higher dimensions binning and KDE fail, so **every multi-dimensional quantity from here on uses Kraskov.**

Parameter side

Kraskov between the representation and the parameters:

- ▶ a alone
- ▶ σ alone
- ▶ $\theta = (a, \sigma)$ together

$\implies MI(S, \theta)$ and $MI(X, \theta)$

Every column is standardised first, otherwise the biggest columns alone would decide who is a neighbour.

Data side

Instead of the 64 raw numbers, five hand-made physical features $\phi(X)$: where the series sits and how big its jumps are.

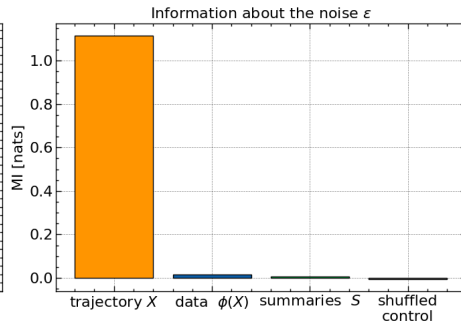
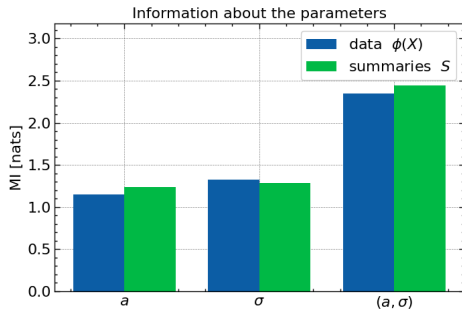
Noise side

The trajectory feels the accumulated kicks:

1. $\text{cumsum}(\varepsilon)$
2. compress to 6 numbers with PCA
3. Kraskov directly

For $MI(X, \varepsilon)$ we use the first 6 PCA components of X ; it's only a lower bound, since the true value is infinite.

The four mutual informations



Parameters: sufficient

The 4 summaries keep as much as the data features.
 $\phi(X)$ is itself a hand-made compression, so S can even come out slightly above it.

Noise: concentrated

The trajectory carries > 1 nat about its own noise.
The summaries ≈ 0.005 , which is at the level of the shuffled control (true value 0).

Is $MI(S, \varepsilon) \approx 0$ a result or an artefact?



The noise is compressed before the estimator sees it,
 $g(\varepsilon) = \text{PCA}_6(\text{cumsum } \varepsilon)$. By data processing,
 $MI(S, g(\varepsilon)) \leq MI(S, \varepsilon) \implies$ we measure a lower bound.

A lower bound alone cannot prove that a quantity is small, but
we also have that:

- ▶ the compressed trajectory gives > 1 nats, meaning that
the **detector is not blind**.
- ▶ shuffling noise give the estimator's floor (true value of 0)
that is on the **same level of shuffled control**.

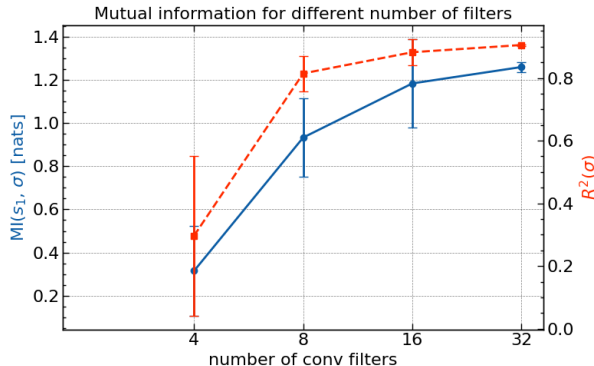
These results show that the estimator's result is not an artefact,
meaning that summaries contain no information about noise.

Trajectory X vs the 6 directions **1.114**

Summaries S , same directions 0.005

|Shuffled control| **≈ 0.006**

values in nats

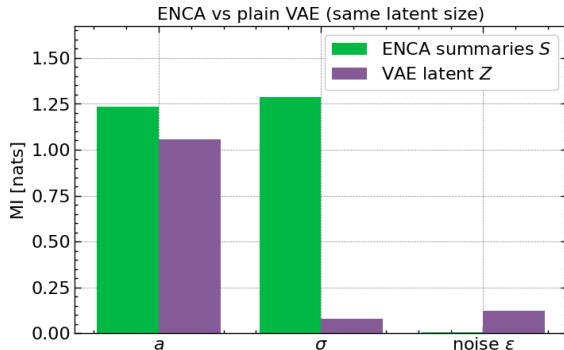


mean $\pm$ spread over 3 trainings (seeds) per width

For each encoder width we retrain the ENCA and measure the information the σ -summary keeps and the accuracy of the σ recovery.

filters	$MI(s_1, \sigma)$	$R^2(\sigma)$
4	0.32 ± 0.21	0.30 ± 0.26
8	0.93 ± 0.18	0.82 ± 0.06
16	1.18 ± 0.20	0.88 ± 0.04
32	1.26 ± 0.02	0.91 ± 0.003

They rise together, and the gain flattens after 16 filters: **mutual information is a usable diagnostic of the compression quality.**



Comparison with a Kingma-Welling **VAE** with the same latent size, but no noise channel and no parameter supervision: only compress and rebuilds x .

nats	ENCA S	VAE Z
$MI(\cdot, a)$	1.24	1.06
$MI(\cdot, \sigma)$	1.29	0.08
$MI(\cdot, \epsilon)$	0.005	0.12
recon. MSE	0.003	0.011

- ▶ Both keep a .
- ▶ VAE throws σ away and stores some noise.

A small jitter barely changes the reconstruction error. So, handing the noise to the decoder is what makes the difference.

Does it depend on the KL weight β ?



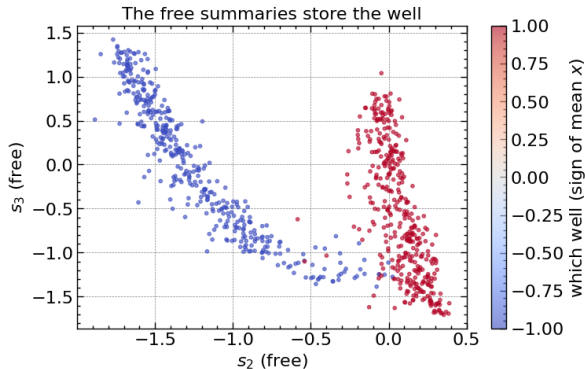
$$\text{Loss} = \text{Reconstruction} + \beta \cdot \text{KL}$$

We want to be sure that the σ loss does not depend on the β coefficient, so we retrain it with smaller β s:

β	recon. MSE	$MI(Z, a)$	$MI(Z, \sigma)$	$MI(Z, \varepsilon)$
1.0	0.0114	1.06	0.08	0.12
0.1	0.0058	0.92	0.09	0.14
0.01	0.0038	0.38	0.15	0.39

A smaller β reconstructs better, but σ stays low (≤ 0.15 nats) while the information about the noise grows (up to 0.39): with more room, the VAE stores the noise, not σ . **The gap with the ENCA comes from the missing noise channel, not from the KL term.**

The free dimensions and the bistability ($q > p$)



test series in the plane of the two free summaries, coloured by well

Two summaries are not enough. The decoder must also know which well the trajectory is in, which is not a parameter: it is decided by the noise during the warm-up.

The well is like a coin: at most $\log 2 = 0.69$ nats.

s_2 carries **0.63** of it, s_0 and s_1 only 0.02: **the network stores it in a spare dimension by itself.**

No contradiction with $MI(S, \varepsilon) \approx 0$. The warm-up noise is never given to the decoder, so the encoder must store the well.

- 1 **The ENCA works on the bistable model.** Both parameters recovered ($R^2 = 0.92$ and 0.94), and the series rebuilt from 4 summaries plus the bare noise.

- 2 **Information theory confirms the compression is selective.** $MI(S, \theta) \approx MI(X, \theta)$ and $MI(S, \varepsilon) \approx 0$, while the series itself carries a lot of information about the noise.

- 3 **Mutual information is a usable diagnostic.** More filters $\rightarrow$ more information about $\sigma \rightarrow$ better inference, flattening after 16.

- 4 **The noise conditioning makes the difference.** A plain VAE of the same size keeps a but loses σ , and stores noise instead.

- 5 **The free dimensions have a physical meaning.** One of them stores the occupied well (almost 1 bit) $\Rightarrow$ the $q > p$ phenomenon of the paper.